Nick Leung EE446 SP26 Lab 5 Report

1. Converting raw audio into spectrogram/MFCC-style features instead of directly using the raw time-domain waveform because Fourier transforms and looking in the frequency domain reduce the computational complexity for a model compared to the time domain. On the Mel scale, equal perceptible differences in pitch correspond to equal distances on the spectrogram, so it's easier for a CNN to distinguish features.
2. The representative dataset was used for post-training quantization. Giving the model realistic input features for quantization can help the quantizer choose the correct scale and zero points.
3. `audio_processor.get_data()` gathers samples from the dataset, applying any necessary transformations. In other words, it converts raw sound data into the format that the model needs.
4. `g_model` is the byte array that represents the quantized model. `g_model_len` is the length (in bytes) of `g_model`.

```
Heard yes (210) @40512ms
Heard yes (206) @42032ms
Heard unknown (206) @43920ms
Heard unknown (221) @45456ms
Heard no (201) @47872ms
```